(8)

5 stage

RISC

IF → Instruction fetch (all instructions)

ID → Instruction decode (all - except writting port not used)

EX → Execution (ALU, MEM, MEM control not used - add

MEM → Memory R - MEM, W - MEM

WB → Write back → ALU e MEM reg e write data

10
9
1 addi x11, x12, 5 → x11 → 22 = 33 old x11
2 add x13, x11, x12 x → x13 → 22+11 old x11
3 add x14, x11, 15 → x14 → 11+15 old x13 26
4 add x15, x11, x11 x15 → 22+22 = 54
ok new x11

x11 → 11
x12 → 22

54

7 After adding x17 with 5, we save the result in x11, however with pipelining the new value of x11 will only be written two instruction ahead so the x11 will contain the old value = 11

7 x13 will sum x12 with x11, however x11 will have the old value 11
so result is 33 (22+11)

3 x14 will also receive x11, that will have old x11, 11 so result in x14 will be 15 + 11 = 26

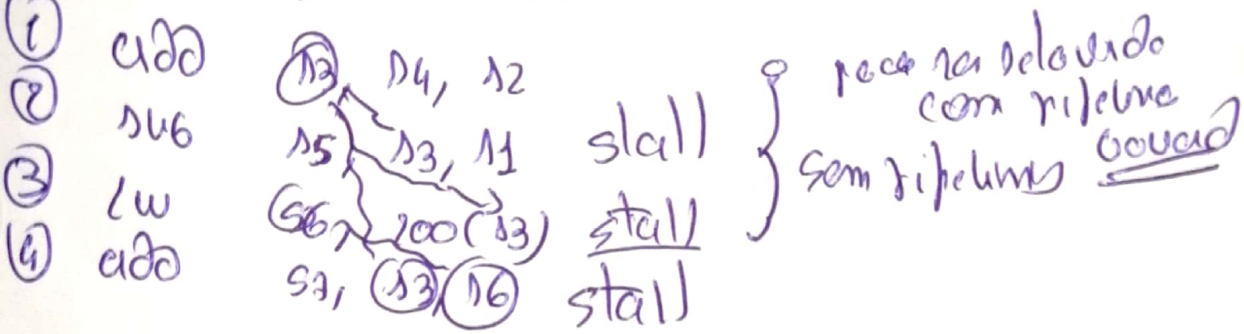
4 x15 will have x11 with x11, in this case x11 from first instruction is already projected to x11 correctly so the value is 22 instead of 11
now x15 is 22 + 22 = 54

Isn't true there defining 12 others in ex

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(10)

Identify the dependencies of the following code and with conflicts can be resolved and with of the generate stall



existem dependencias entre as tres ultimas e a primeira ura $3 que poder ser resolvido com atalho "forward" na ritelne, a ultima instrcap $add $7, $3, $6 tambem depende da instrcap lw. $6, 200($3) usando atalho com nop necessarios

O ritelnes com bouad sem atalho alguns problem

② and ③ can be solved by forwarding while ④ always needs to stall we can change ③ to second place and avoid stall ④ instruction

① e ② resolvido atalho MEM - MEM

③ ④ stall one cicle since lw only gets available in stage Mem Uma bourding Mem - MEM only requires 1 cicle e otherwise 2 cicles were needed

(4)

$N_{instr} = 10^3$

lw
add
lw
add
sw
add
:

a) CPI with forwarding (delay 1 cycle only Hem-Mul

$CPI = \frac{N\ cycles}{N\ instructions} = \frac{4+2}{4} = 1,5$ stall
6 cycle

b) CPI without forwarding

(delay 2 cycles lw → sw)

$CPI = \frac{N\ cycles}{N\ inst} = \frac{1\times3 + 1\times3}{4} + 1\times3 + 1\times3$

$= \frac{1\times(3) + 1\times3 + 1\times3 + 1\times3}{4}$ 1 + 3 + 1 13 + 1 13 + 1 13

$= \frac{k}{4} = 3$ → stall 2 cycles between lw and add on stall 2 cycles add and next sw

(12)

IF	ID	EX	MEM	WB
300ps	400ps	350ps	500ps	100ps

9) qual o periodo minimal for a path with and without pipelining

Unique cicle = Σ all time slots
Multi cicle = max (all time slots)
Pipelining = max (all tim slots)

Unicicle time = 300ps + 400ps + 350ps + 500ps + 100ps = 1650ps

Multicicle = max (300ps, 400ps, 350ps, 500ps, 100ps)
= 500ps

Pipelining = max (300ps, 400ps, 350ps, 500ps, 100ps)
= 500ps → ~~total~~ = 5 slots × 500ps =>

6) with is the latency total of lw load instruction in pipeline with and without pipelining

without pipelining
= 1650ps

Note: load (lw) requires 5 stages

with pipeling (we have 5 stages, some instruction will only need 3 or 4, but in pipelining all instruction requiring pass 5 stages even not doing nothing

(13) a. -

So the latency with refreshing
is 5 x 300ps = 2500ps all net s
stages



c) e)

g

f

(

(3) ① lw ①, 40(t6)
② add t6, t2, t2
③ sw t6, 50(t1)

a) Indicate all dependencies

① and ③ lw depends on t1 from 1st instruction

c) Assume the cpu (Pipeline) without forwarding, indicate conflict and where nop must set in here

Regarding ③ ② and 1 generate conflicts so without forwarding to ensure value consistency we need to add the nop between add and lw

lw ①, 40(t6)
add t6, t2, t2
nop
nop
sw t6, 50(t1) → ④

c) Assuming cpu implements all forwarding mw-mw and alu-mw indicate the conflicts and when nop must be placed

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e)

with MEM - WB and MEM - WB

lw t5, 40(t6)
add t6, t4, t2
sw t6, 80(s5)

} from - MEM

1 cycle OK

d) Total time with forwarding and no forwarding

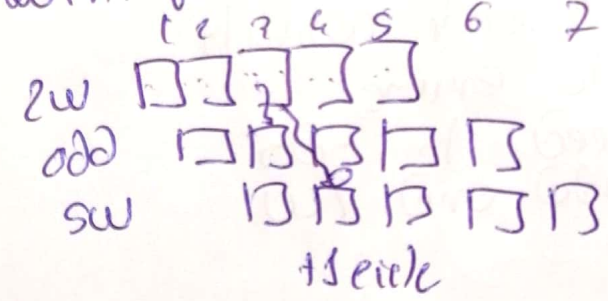
Assume 100 ns times

No forwarding 300 ns

all forwarding 400 ns

only MEM - WB forwarding 3600 ns

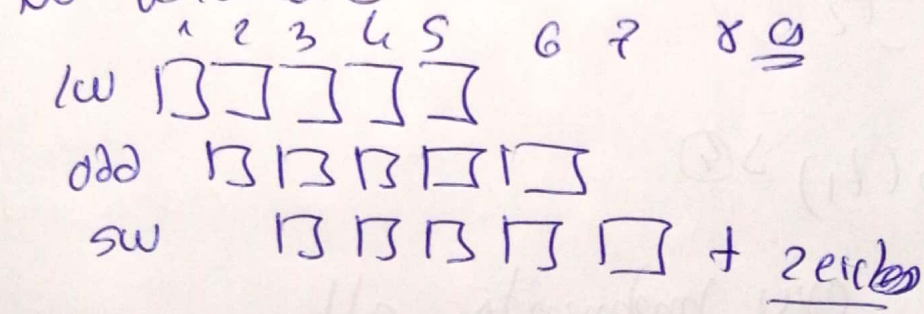
with forwarding



7 cycle

7 × 400 ns = 2800 ns

No forwarding



+ 2 cycles

9 × 300 ns = 2700 ns

Speed up = 2700 / 2800 = 0.96

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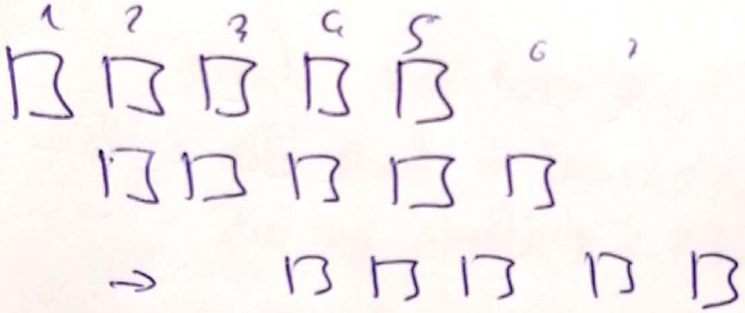
e)

Only loads NOP NOP
we only need 2 cicles between lw and sw
so

```
lw t1 40(16)
nop
nop
add t6, t2, t1
sw t6, 80(t1)
```

f) Determine total time to execute the sequence with only NOP-NOP between dependent instructions without forwarding

NOP - NOP



12 cicles

Com NOP - NOP

Clock = 8 × 360 = 2880 ns

Without NOPs

Clock =

8 × 300 = 2700 ns

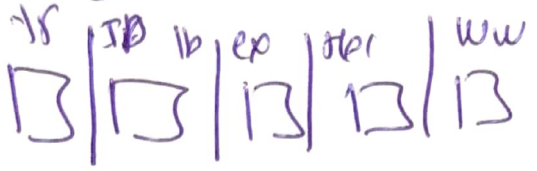
Speedup = 2700 / 2880 = 0.93

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(14)

lw $2, 40(16)

$t6 = 200



IF | ID → PC + 4

ID | EX → PC + 4, Read data 1 = 200
Read data 2 = ?, Imm 40 - endereço

Sinais controlo: RWSrc = 1, RW OP = 00
MemRead = 0, MemRead = 1, MemWrite = 0
MemtoReg = 1 RegWrite = 1

EX | MEM

Sinal Add Sum = PC + 4,
ALU Result = 200 + 40 = 240
Zero = 0, Read data 2 = ?, Resultado ALU = 240

Sinais controlo: MemRead = 0, MemRead = 1, MemWrite = 0
MemtoReg = 1, RegWrite = 1

MEM / WB

Read data = valor contido 240
no registo destino ($2 ⇒ 07

Sinais controlo: MemtoReg = 1, RegWrite = 1

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(15) g) What are the activities associated with this instruction in stage EX/MEM

EX: Compute the address from where to read

MEM: Read from memory the data.

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(5)

```
loop:
    lw   t1, 40(t6)
    add  t5, t5, t3
    add  t6, t6, t3
    sw   t1, 20(t3)
    beq  t1, zero loop
```

Iter	Instr	pipeline				
2	add	WB				
2	add	(MEM)	WB			
2	sw	EX	MEM	(WB)		
2	beq	ID	EX	MEM	(WB)	
3	lw	IF	ID	EX	MEM	WB
3	add		IF	ID	EX	(MEM)
3	add			IF	ID	EX
3	sw				IF	ID
3	beq					IF

6) Pseudo which is contain IF/ID ?

contain the value of last beq instruction code and the instruction code address lw

c) only 1 instruction uses all cycles

1 one lole of 5 instruction no 20%

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16

① add $t2, $s1, 12
② addi $t3, $t2, -15
③ lw $t5, 12($t2)
④ or $t5, $t5, $t3
⑤ lw $s1, 0($t3)

dependencies

dependency

forward = MEM - MEM
MEM - MEM

a) with dependencies and how can be solved forwards to minimize execution time

solution ① and ② via $t2 → solved forwards MEM - MEM

solution ① and ② via $t2 → solved forwards MEM - MEM (1 cycle)

solution ③ and ④ via $t5 → need nop and MEM - MEM forward (3 without forward)

solution 2 and 4 via $t3 no conflicts only nop instructions 3 and 4

solution 2 and 5 via $t3 → all ok

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6)

IF ID EX MEM WB
300ps 400ps 350ps 500ps 100ps

~~Pipeline max or throughput~~

Pipeline rate or throughput

$\Rightarrow \max(IF, ID, EX, MEM, WB)$

$= 500ps$

we have $\boxed{\text{Texe} = \text{Nstages} \times T}$

$\text{Tex} = \text{Nstages} \times 500 \times 10^{-12}$

1 2 3 4 5 6 7 8 9 10

Nstages $= 10$ $\quad$ Texe $= 10 \times 500 \times 10^{-12}$ s

e) Consider the code segment execute 10^6 consecutive times. determine CPI

Considering loop we need to consider to delay ③ or ④ and now 5 and 1 in the loop

$$CPI = \frac{5+2 \text{ delays}}{5} = 1.4$$

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(17)

```
sw    x29 , 11(x16)
lw    x19 , 8(x16)    ) - conflit
DUG   x17, x15, x14
beq   x17, x0, L
add   x15, x11, x14
DUG   x15, x30, x14
```

9)

sub (1) x17.

```
                IF  ID  EX  MEM  WB
                    IF  ID  EX   MEM  WB
                        IF  ID   EX   MEM  WB
                            XXX  XXX  IF   ID   EX MEM WB
```

```
SUB (x17), x15
beq (x17) → delay (2)
```

6)

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